Peiyuan Zhai

Ph.D. Candidate in Electrical Engineering · Signal Processing Systems, TU Delft

Delft, The Netherlands | (+31) 06-83104008 | hantyou.zhai@gmail.com

zhai.slipzhai.cc | github.com/hantyou | linkedin.com/in/peiyuan-zhai

Education

Ph.D. in Electrical Engineering — Feb 2023 – Feb 2027 (exp.)
Delft University of Technology · Signal Processing Systems Group — *Delft, NL*

M.Sc. in Electrical Engineering — Sep 2020 – Aug 2022
Delft University of Technology · Faculty of EEMCS · GPA 8.32 / 10 — *Delft, NL*

B.Eng. in Information Engineering — Sep 2016 – Jul 2020
Xi'an Jiaotong University · School of Information and Communications Engineering · GPA 86.44 / 100 — *Xi'an, CN*

Publications

Journal Articles

[J3] **P. Zhai**, G. Joseph, N. J. Myers, and A. Pandharipande, "Camera-Aided Binary Prior Support-Informed Occupancy Grid Mapping," *IEEE Sensors Journal*, vol. 26, no. 3, pp. 4340–4348, Feb. 2026. doi:10.1109/JSEN.2025.3642255

[J2] F. Harraway, **P. Zhai**, G. Joseph, and A. Pandharipande, "Accelerated Pattern-Coupled Sparse Bayesian Learning for Automotive Occupancy Mapping," *IEEE Sensors Journal*, vol. 25, no. 22, pp. 41801–41810, Nov. 2025. doi:10.1109/JSEN.2025.3617133

[J1] **P. Zhai**, G. Joseph, N. J. Myers, C. Onen, and A. Pandharipande, "Spatial Sparsity-Aware Radar-LiDAR Fusion for Occupancy Grid Mapping in Automotive Driving," *IEEE Sensors Journal*, vol. 25, no. 17, pp. 33328–33338, Sep. 2025. doi:10.1109/JSEN.2025.3592023

Conference Papers

[C4] **P. Zhai**, M. Dai, G. Joseph, N. J. Myers, and A. Pandharipande, "Radar-Based Sparsity-Aware Dynamic Occupancy Grid Mapping," in *2026 IEEE SENSORS*, Oct. 2026. conference paper

[C3] F. Harraway, **P. Zhai**, A. Pandharipande, and G. Joseph, "Computationally-Efficient Sparsity-Aware Occupancy Grid Mapping for Automotive Driving," in *2025 IEEE SENSORS*, 2025. doi:10.1109/SENSORS59705.2025.11331194

[C2] **P. Zhai**, G. Joseph, N. J. Myers, C. Onen, and A. Pandharipande, "Sparsity-Aware Occupancy Grid Mapping for Automotive Driving Using Radar-LiDAR Fusion," in *2024 IEEE SENSORS*, Oct. 2024, pp. 1–4. doi:10.1109/SENSORS60989.2024.10785054

[C1] **P. Zhai** and R. T. Rajan, "Distributed Gaussian Process Hyperparameter Optimization for Multi-Agent Systems," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, Jun. 2023, pp. 1–5. doi:10.1109/ICASSP49357.2023.10096267

Patent Applications

[P1] **P. Zhai**, A. Pandharipande, G. Joseph, and N. J. Myers, "Occupancy Grid Mapping System and Method." Published Oct. 2025 as US20250336088A1 (30 Oct.), EP4641250A1 (29 Oct.), and CN120846354A (28 Oct.).

Doctoral Research

Doctoral Research on Autonomous Perception: Occupancy Grid Mapping — Feb 2023 – Feb 2027 (exp.)
Signal Processing Systems, Delft University of Technology · Supervised by Dr. Geethu Joseph — *Delft, NL*

- Studied camera-aided sparsity-aware occupancy grid mapping (OGM). Proposed a novel algorithm that incorporates machine-learning-based visual detections into the prior distribution of the sparse Bayesian learning OGM framework. Published a journal paper [J3] and a patent [P1] on this topic.
- Proposed a fusion strategy under the sparsity-aware OGM framework that softly fuses LiDAR and radar measurements for robust OGM performance. Published a conference paper [C2] and a journal paper [J1].
- Proposed a dynamic OGM algorithm that exploits dynamic radar measurements across time frames [C4]. Currently extending this work to more information sources and incorporation strategies.
- Daily supervisor of an M.Sc. thesis on computationally efficient pattern-coupled sparse Bayesian learning for occupancy mapping [J2, C3]. Evaluated methods on the nuScenes and RADIATE datasets.

Skills

Research	Occupancy grid mapping, autonomous perception, multi-sensor fusion, sparse Bayesian learning, distributed signal processing
Methods	Bayesian inference, convex and non-convex optimization, Gaussian processes, proximal ADMM
Sensors	Automotive radar, LiDAR, camera detections; nuScenes and RADIATE datasets
Tools	Python, C/C++, MATLAB, Linux, Git, LaTeX, FPGA
Languages	English (fluent), Chinese (native), Dutch (beginner)

Research Profile

Ph.D. candidate in Electrical Engineering at Delft University of Technology, working on probabilistic perception and occupancy grid mapping for autonomous systems. Research interests include radar–LiDAR–camera fusion, sparse Bayesian learning, dynamic-scene mapping, and robust perception under challenging sensing conditions. Published in IEEE Sensors Journal, IEEE SENSORS, and ICASSP; co-inventor on published patent applications in occupancy grid mapping; supervised an M.Sc. thesis that led to two co-authored publications.

Supervision

Daily Supervisor, M.Sc. Thesis of Frank Harraway Graduated Sep 2025

Signal Processing Systems, Delft University of Technology · With Dr. G. Joseph (TU Delft) and Dr. A. Pandharipande (NXP) *Delft, NL*

- Thesis: "Computationally-Efficient Sparsity-Aware Occupancy Grid Mapping for Automotive Driving", M.Sc. Electrical Engineering, TU Delft, 2025.
- Guided the student's work on reducing the computational cost of pattern-coupled sparse Bayesian learning for occupancy mapping, evaluated on nuScenes and RADIATE.
- The thesis led to a journal article [J2] and a conference paper [C3], each co-authored as second author.

Earlier Research

M.Sc. Thesis on Distributed Gaussian Processes Dec 2021 – Aug 2022

Circuits and Systems, Delft University of Technology · Supervised by Dr. Raj Thilak Rajan · Score 8.5 / 10 *Delft, NL*

- Researched Gaussian Process (GP) based unknown-field learning in multi-agent systems and simulated state-of-the-art algorithms.
- Proposed two fully-distributed asynchronous proximal ADMM algorithms for GP hyperparameter optimization, extending the state of the art to fully-distributed systems without a fusion center.
- Proposed several fully-distributed GP aggregation strategies for consistent and accurate unknown-field prediction. Published at ICASSP 2023 [C1].

B.Eng. Thesis on Video Background Modeling Nov 2019 – Jun 2020

Multi-media Processing Group, Xi'an Jiaotong University · Supervised by Assoc. Prof. Ruiping Qiao *Xi'an, CN*

- Researched state-of-the-art background modeling for target tracking.
- Proposed a selective-update strategy for Gaussian Mixture Model (GMM) background modeling and evaluated it on real-world videos.
- Designed a routine for implementing the proposed algorithm on Digital Signal Processor platforms.

Presentations

Oct 2026 **Tutorial** (co-presented with Dr. Ashish Pandharipande, NXP Semiconductors, and Dr. Geethu Joseph, TU Delft): "From Sensor Measurements to Environment Perception: Occupancy Maps for Autonomous Driving and Robotics," IEEE SENSORS 2026, Rotterdam, the Netherlands. Slides

Sep 2026 "Sparsity-Aware Occupancy Grid Mapping," 3 Minute Thesis, EUSIPCO 2026, Bruges, Belgium. Video

Personal

Interests Cycling, cooking, photography